

2019-Ph-H1-Q1

Section: Our Dynamic Universe

Topic: Motion, Equations and Graphs

Question:

A car accelerates from rest with constant acceleration. Which of the following velocity–time graphs represents this motion?

Answer:

The correct graph is a straight line **starting at the origin** and **sloping upwards** — this indicates increasing velocity with time, i.e. constant acceleration.

Correct Option: A

Guidance for Students:

Constant acceleration means equal increases in velocity over equal time intervals, giving a straight sloped line.

Revision Tips:

- A **straight diagonal line** on a v – t graph means constant acceleration.
- A **flat line** on a v – t graph means constant velocity.
- The **area under a v – t graph** gives displacement.